



Unit-1
class-1

Unit-1 class-1

1. $\int_1^2 \int_3^4 (xy + e^y) dy dx$ ans: $\frac{21}{4} + e^4 - e^3$

A. $\int_1^2 \left(\frac{xy^2}{2} + e^y \right)_3^4 dx = \int_1^2 \left(\frac{16x}{2} + e^4 - \frac{9x}{2} - e^3 \right) dx$
 $= \int_1^2 \left(\frac{7x}{2} + e^4 - e^3 \right) dx = \left[\frac{7x^2}{4} + x(e^4 - e^3) \right]_1^2 = 7 + 2(e^4 - e^3) - \frac{7}{4} - (e^4 - e^3)$
 $= \frac{21}{4} + e^4 - e^3$

2. $\iint (x+y)^2 dx dy$ over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ans: $\frac{1}{4} \pi ab (a^2 + b^2)$

A. Assume, $u = \frac{x}{a}$ & $v = \frac{y}{b} \Rightarrow a du = dx$ & $b dv = dy$

$I = \iint_R (au + bv)^2 \cdot (ab du dv)$
 $= ab \iint_R (a^2 u^2 + b^2 v^2 + 2abuv) du dv \Rightarrow$ bounded by $u^2 + v^2 = 1$

Also, $\iint_R u^2 du dv = \iint_R v^2 du dv$ & $\iint_R uv du dv = 0$

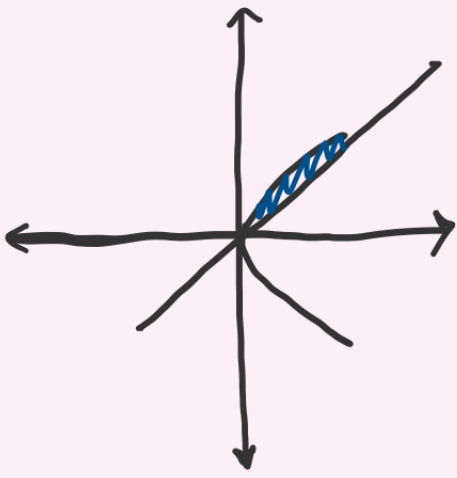
$I = ab \left(a \iint_R u^2 du dv + 2ab \iint_R uv du dv + b \iint_R v^2 du dv \right)$
 $= ab(a+b) \iint_R u^2 du dv$

Assume some $J = \iint_R u^2 du dv = \iint_R v^2 du dv$

Then, $J = \frac{1}{2} \iint_R (u^2 + v^2) du dv = \frac{1}{2} \int_0^{2\pi} \int_0^1 r^2 (r dr d\theta)$
 $= \frac{1}{2} \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^1 d\theta = \frac{1}{8} [\theta]_0^{2\pi} = \frac{\pi}{4}$

$I = \frac{\pi ab(a+b)}{4}$

3. $\iint xy(x+y) dx dy$ over the area between $y = x^2$ and $y = x$ ans: $\frac{3}{56}$



l^{el} to y-axis: $y: x^2$ to x
 $x: 0$ to 1

$\int_0^1 \int_{x^2}^x xy(x+y) dy dx$

$= \int_0^1 \left(\frac{x^2 y^2}{2} + \frac{xy^3}{3} \right)_{x^2}^x dx$

$= \int_0^1 \left(\frac{x^4}{2} + \frac{x^4}{3} - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$

$= \int_0^1 \left(\frac{5x^4}{6} - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$

$= \left[\frac{5x^5}{30} - \frac{x^7}{14} - \frac{x^8}{24} \right]_0^1$

$= \frac{5}{30} - \frac{1}{14} - \frac{1}{24}$

$= \frac{3}{56}$

(01) $I = \iint_A (x+y)^2 dx dy = \iint_A (x^2 + y^2 + 2xy) dx dy$

$x \rightarrow -a\sqrt{1 - \frac{y^2}{b^2}}$ to $a\sqrt{1 - \frac{y^2}{b^2}}$
 $y \rightarrow -b$ to b
 $I = \int_{-b}^b \int_{-a\sqrt{1 - \frac{y^2}{b^2}}}^{a\sqrt{1 - \frac{y^2}{b^2}}} (x^2 + y^2 + 2xy) dx dy$

$\iint 2xy = 0$ coz $a^2 \left(1 - \frac{y^2}{b^2}\right) - a^2 \left(1 - \frac{y^2}{b^2}\right) = 0$

V

$= 2 \times 2 \int_0^b \int_0^{a\sqrt{1 - \frac{y^2}{b^2}}} (x^2 + y^2) dx dy$

$= 4 \int_0^b \left[\frac{x^3}{3} + xy^2 \right]_0^{a\sqrt{1 - \frac{y^2}{b^2}}} dy$

$= 4 \int_0^b \left(\frac{a^3}{3} \left(1 - \frac{y^2}{b^2}\right)^{3/2} + ay^2 \sqrt{1 - \frac{y^2}{b^2}} \right) dy$

$\frac{y}{b} = \sin t \Rightarrow dy = b \cdot \cos t \cdot dt$

$y = 0 \Rightarrow t = 0$; $y = b \Rightarrow t = \pi/2$

$= 4 \int_0^{\pi/2} \left(\frac{a^3}{3} (1 - \sin^2 t)^{3/2} \cdot b \cdot \cos t dt + a b^2 \sin^2 t \cdot \cos t \cdot b \cos t dt \right)$

$= 4 \int_0^{\pi/2} \left(\frac{a^3 b}{3} \cos^4 t dt + \frac{ab^3}{4} \sin^2 t \cdot \cos^2 t dt \right)$

[Solved using calculator]

$= 4 \left(\frac{a^3 b}{3} \left(\frac{\sin 4t + 8 \sin 2t + 12t}{32} \right) \Big|_0^{\pi/2} + \frac{ab^3}{4} \cdot \left(\frac{-\sin 4t - 4t}{32} \right) \Big|_0^{\pi/2} \right)$

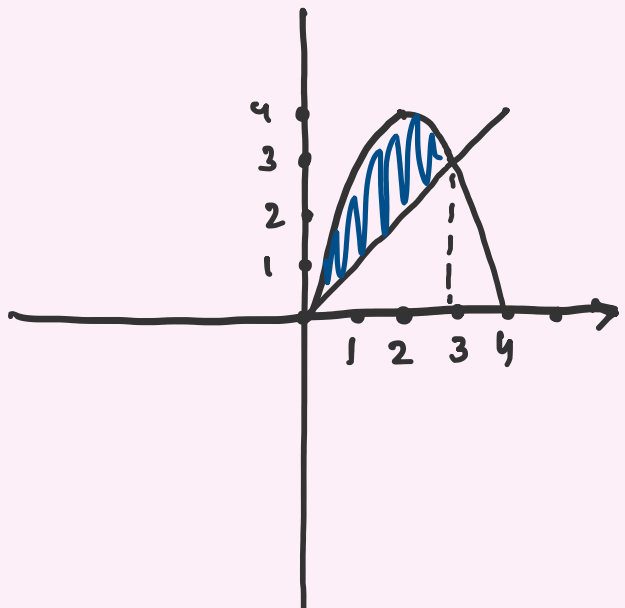
$= 4 \left(\frac{a^3 b}{3} \cdot \frac{\pi}{16} + \frac{ab^3}{4} \cdot \frac{\pi}{4} \right)$

$= \frac{4ab\pi}{16} (a^2 + b^2) = \frac{\pi ab(a^2 + b^2)}{4}$

Unit-1 class-2

1. Find by double integration, the area lying between the parabola $y = 4x - x^2$ and the line $y = x$
ans: 4.5

A.



$$y = 4x - x^2 \qquad y = x$$

$x = 0$	$y = 0$
$x = 1$	$y = 3$
$x = 2$	$y = 4$
$x = 3$	$y = 3$

Parallel to x -axis

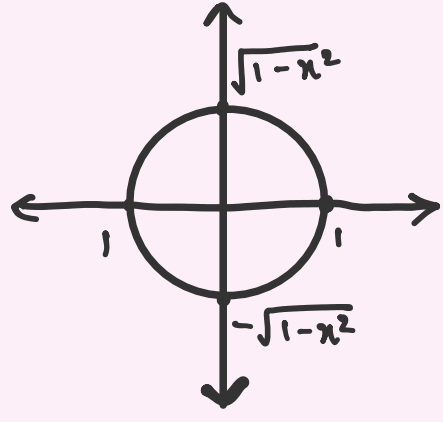
$y: x \text{ to } 4x - x^2$

$x: 0 \text{ to } 4$

$$\int_0^3 \int_x^{4x-x^2} dy dx = \int_0^3 (4x - x^2 - x) dx = \int_0^3 (3x - x^2) dx$$
$$= \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_0^3 = \frac{27}{2} - 9 = \frac{9}{2} = 4.5$$

2. Find the volume bounded by the xy -plane, the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 3$
ans: 3π

A.



$$x + y + z = 3$$
$$z = 3 - x - y$$
$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (3 - x - y) dy dx$$
$$= \int_{-1}^1 \left[3y - xy - \frac{y^2}{2} \right]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dx$$
$$= \int_{-1}^1 \left[3\sqrt{1-x^2} - (-3\sqrt{1-x^2}) - (x\sqrt{1-x^2} - (-x\sqrt{1-x^2})) - \frac{(1-x^2) - (1-x^2)}{2} \right] dx$$
$$= \int_{-1}^1 (6\sqrt{1-x^2} - 2x\sqrt{1-x^2}) dx$$

①

②

For ①, $\int_{-1}^1 \sqrt{1-x^2} dx$

$$x = \sin \theta \quad \left| \begin{array}{l} x=1, \quad \theta = \pi/2 \\ x=-1, \quad \theta = -\pi/2 \end{array} \right.$$
$$dx = \cos \theta d\theta$$
$$\Rightarrow \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = \int_{-\pi/2}^{\pi/2} \left(\frac{\cos 2\theta + 1}{2} \right) d\theta = \frac{\pi}{2}$$

For ②, $\int_{-1}^1 2x\sqrt{1-x^2} dx$

$$u = 1 - x^2$$
$$du = -2x dx$$
$$\int_0^0 \sqrt{u} du = 0$$

①

-②

$$6\left(\frac{\pi}{2}\right) - 0 = 3\pi //$$

3. Find the average value of the function e^{x+y} over the region $R = [0,2] \times [0,2]$ ans: $\frac{(e^2-1)^2}{4}$

A.

$$\int_0^2 \int_0^2 e^{x+y} dy dx = \int_0^2 [e^y]_0^2 e^x dx = \int_0^2 (e^2 - 1) e^x dx = (e^2 - 1)^2$$

Now,

$$\int_0^2 \int_0^2 dy dx = \int_0^2 (2 - 0) dx = \int_0^2 2 dx = 2 \times (2 - 0) = 4$$

Average value = $\frac{(e^2 - 1)^2}{4}$

Unit-1 class-3

1. $\int_0^a \int_y^a \frac{xdxdy}{x^2 + y^2}$

ans: $\frac{\pi a}{4}$

x varies from y to a

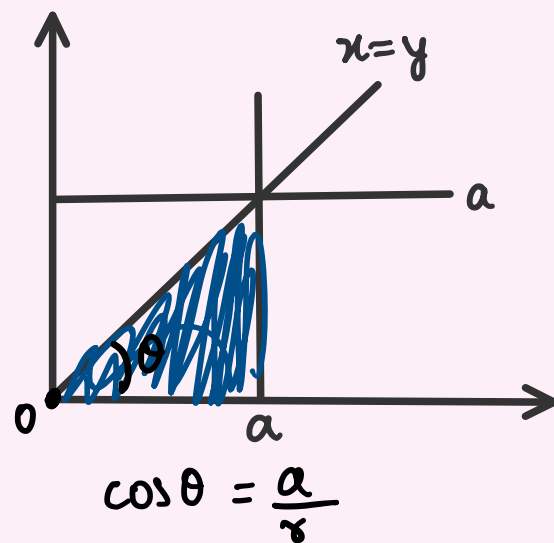
y varies from 0 to a

θ varies from 0 to $\pi/4$

r varies from 0 to $a \sec \theta$

$$\int_0^{\pi/4} \int_0^{a \sec \theta} \frac{r \cos \theta \cdot r dr d\theta}{r^2}$$

$$= \int_0^{\pi/4} [r]_0^{a \sec \theta} \cos \theta d\theta = a \sec \theta \cdot \cos \theta \int_0^{\pi/4} d\theta = \frac{a \pi}{4}$$



2. $\int_0^a \int_y^a \frac{x^2}{\sqrt{x^2 + y^2}} dx dy$

ans: $\frac{a^3}{3} \log(\sqrt{2} + 1)$

x varies from y to a

y varies from 0 to a

r varies from 0 to $a \sec \theta$

θ varies from 0 to $\pi/4$

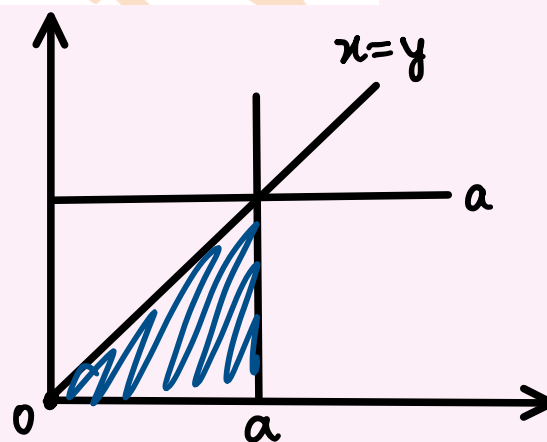
$$\int_0^{\pi/4} \int_0^{a \sec \theta} \frac{r^2 \cos^2 \theta}{r} dr d\theta$$

$$= \int_0^{\pi/4} \left[\frac{r^3}{3} \right]_0^{a \sec \theta} \cos^2 \theta d\theta$$

$$= \frac{a^3}{3} \int_0^{\pi/4} \sec \theta d\theta$$

$$= \frac{a^3}{3} \left[\log(\sec \theta + \tan \theta) \right]_0^{\pi/4}$$

$$= \frac{a^3}{3} \log(\sqrt{2} + 1)$$



$$= \pi \frac{x^2}{8} = \frac{\pi a^2}{8}$$

Unit 1 class-4

1. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{(1+x^2+y^2)^{3/2}} dx dy$ ans: 2π

$$x = r \cos \theta \quad y = r \sin \theta$$

$\Rightarrow$ All 4 quadrants are considered,

$$\text{So, } \int_0^{2\pi} \int_0^{\infty} \frac{1}{(1+r^2)^{3/2}} \cdot r dr d\theta$$

$$t = 1+r^2$$

$$dt = 2r dr \Rightarrow \frac{dt}{2} = r dr$$

$$\Rightarrow \int_0^{2\pi} \int_1^{\infty} t^{-3/2} \frac{dt}{2} d\theta$$

$$= \int_0^{2\pi} -[t^{-1/2}]_1^{\infty} d\theta$$

$$= \int_0^{2\pi} d\theta = [\theta]_0^{2\pi} = 2\pi$$

2. $\iint y^2 dx dy$ over the area outside $x^2 + y^2 - ax = 0$ and inside $x^2 + y^2 - 2ax = 0$ ans: $\frac{15\pi a^4}{64}$

$$x = r \cos \theta \quad y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$x^2 + y^2 = ax \Rightarrow r^2 = ar \cos \theta$$

$$r = a \cos \theta$$

$$x^2 + y^2 = 2ax \Rightarrow r^2 = 2ar \cos \theta$$

$$r = 2a \cos \theta$$

θ varies from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$

$$\int_{-\pi/2}^{\pi/2} \int_{a \cos \theta}^{2a \cos \theta} r^2 \sin^2 \theta \cdot r \cdot dr d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \left[\frac{r^4}{4} \right]_{a \cos \theta}^{2a \cos \theta} \cdot \sin^2 \theta d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \frac{16-1}{4} \times a^4 \cos^4 \theta \cdot \sin^2 \theta d\theta$$

$$= \frac{15a^4}{4} \int_{-\pi/2}^{\pi/2} (\sin \theta \cos \theta)^2 \left(\frac{1+\cos 2\theta}{2} \right) d\theta$$

$$= \frac{15a^4}{4} \int_{-\pi/2}^{\pi/2} \frac{\sin^2 2\theta}{4} \times \left(\frac{1+\cos 2\theta}{2} \right) d\theta$$

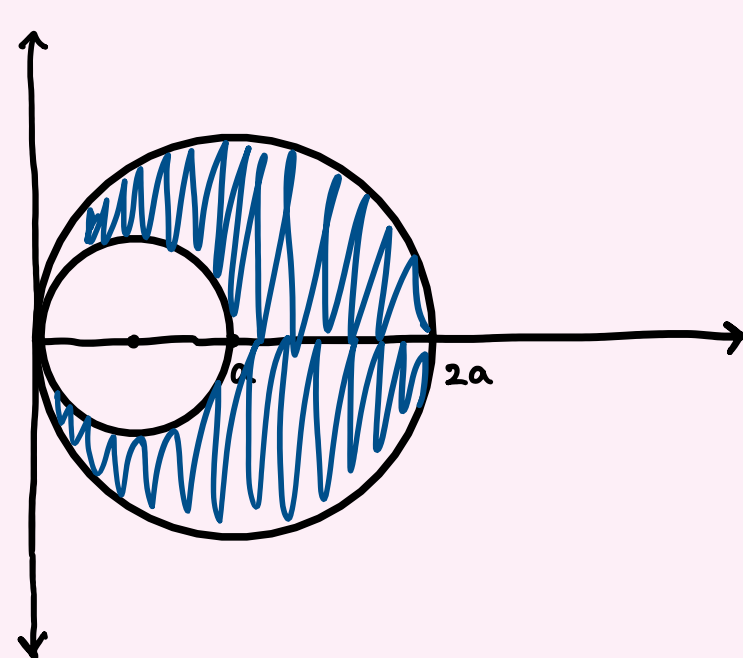
$$= \frac{15a^4}{32} \times 2 \int_0^{\pi/2} \left(\frac{1-\cos 4\theta}{2} \right) (1+\cos 2\theta) d\theta$$

$$= \frac{15a^4}{32} \int_0^{\pi/2} [1 - \cos 4\theta + \cos 2\theta - \cos 4\theta \cdot \cos 2\theta] d\theta$$

$$= \frac{15a^4}{32} \int_0^{\pi/2} [1 - \cos 4\theta + \cos 2\theta - \cos 4\theta \cdot \cos 2\theta] d\theta$$

$$= \frac{15a^4}{32} \left[\theta - \frac{\sin 4\theta}{4} + \frac{\sin 2\theta}{2} - \frac{\sin 6\theta + 3\sin 2\theta}{12} \right]_0^{\pi/2}$$

$$= \frac{15a^4}{32} \times \frac{\pi}{2} = \frac{15\pi a^4}{64}$$



3. $\iint \frac{1-x^2-y^2}{1+x^2+y^2} dx dy$ the integral being extended over all positive values of x and y subject to $x^2 + y^2 \leq 1$. ans: $\frac{\pi^2}{8} - \frac{\pi}{4}$

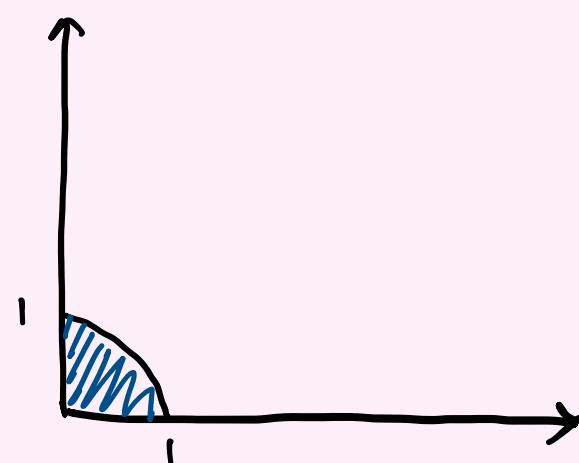
$$x = r \cos \theta, \quad y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$x^2 + y^2 = 1 \Rightarrow r^2 = 1 \Rightarrow r = 1$$

$\hookrightarrow$ No -ve cuz only +ve values

$$x^2 + y^2 = 0 \Rightarrow r = 0$$



$$\int_0^{\pi/2} \int_0^1 \frac{\sqrt{1-r^2}}{\sqrt{1+r^2}} \cdot r dr d\theta$$

$$u^2 = 1+r^2$$

$$\frac{1}{2} du = \frac{1}{2} r dr$$

$$\int_0^{\pi/2} \int_1^{\sqrt{2}} \frac{\sqrt{2-u^2}}{u^2} \cdot u du d\theta$$

$$= \int_0^{\pi/2} \int_1^{\sqrt{2}} \frac{\sqrt{2-u^2}}{u} \cdot du d\theta$$

$$= \int_0^{\pi/2} \left[\frac{u}{2} \sqrt{2-u^2} + \frac{2}{\sqrt{2}} \sin^{-1} \left(\frac{u}{\sqrt{2}} \right) \right]_1^{\sqrt{2}} d\theta$$

$$= \int_0^{\pi/2} \left[0 - \frac{1}{2} + \sin^{-1}(1) - \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right] d\theta$$

$$= \int_0^{\pi/2} \left(\frac{\pi}{2} - \frac{\pi}{4} - \frac{1}{2} \right) d\theta = \int_0^{\pi/2} \left(\frac{\pi}{4} - \frac{1}{2} \right) d\theta$$

$$= \frac{\pi}{4} \left[\theta \right]_0^{\pi/2} - \left[\frac{\theta}{2} \right]_0^{\pi/2}$$

$$= \frac{\pi^2}{8} - \frac{\pi}{4}$$

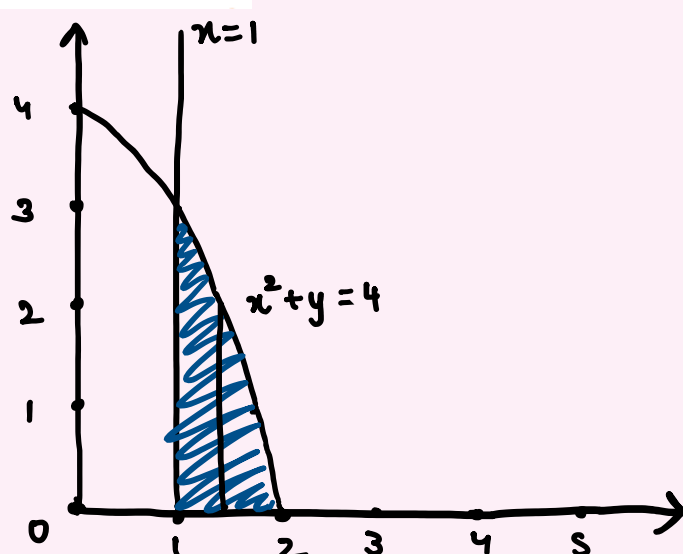
1. $\int_0^3 \int_1^{\sqrt{4-y}} (x+y) dx dy$

ans: $\frac{241}{60}$

x varies from 1 to $\sqrt{4-y}$
 y varies from 0 to 3

||^{el} to y axis,

y varies from 0 to $4-x^2$
 x varies from 1 to 2



$$\begin{aligned} \int_1^2 \int_0^{4-x^2} (x+y) dy dx &= \int_1^2 \left[xy + \frac{y^2}{2} \right]_0^{4-x^2} dx = \int_1^2 \left(x(4-x^2) + \frac{(4-x^2)^2}{2} \right) dx \\ &= \int_1^2 \left(4x - x^3 + \frac{16 + x^4 - 8x^2}{2} \right) dx = \left[2x^2 - \frac{x^4}{4} + 8x + \frac{x^5}{10} - \frac{4x^3}{3} \right]_1^2 \\ &= \left(8 - 4 + 16 + \frac{16}{5} - \frac{32}{3} \right) - \left(2 - \frac{1}{4} + 8 + \frac{1}{10} - \frac{4}{3} \right) = \frac{241}{60} \end{aligned}$$

2. $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$

ans: $1 - \frac{1}{\sqrt{2}}$

y varies from x to $\sqrt{2-x^2}$
 x varies from 0 to 1

||^{el} to x axis,

For R_1 ,

x varies from 0 to $\sqrt{2-y^2}$
 y varies from 1 to $\sqrt{2}$

For R_2 ,

x varies from 0 to y
 y varies from 0 to 1

$$R_1 + R_2 \Rightarrow \int_0^1 \int_0^y \frac{x}{\sqrt{x^2+y^2}} dy dx + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}} dx dy$$

$$t = x^2 + y^2$$

$$\frac{dt}{2} = x dx$$

$$x=0 \Rightarrow t=y^2$$

$$x=y \Rightarrow t=2y^2$$

$$x=0 \Rightarrow t=y^2$$

$$x=\sqrt{2-y^2} \Rightarrow t=2$$

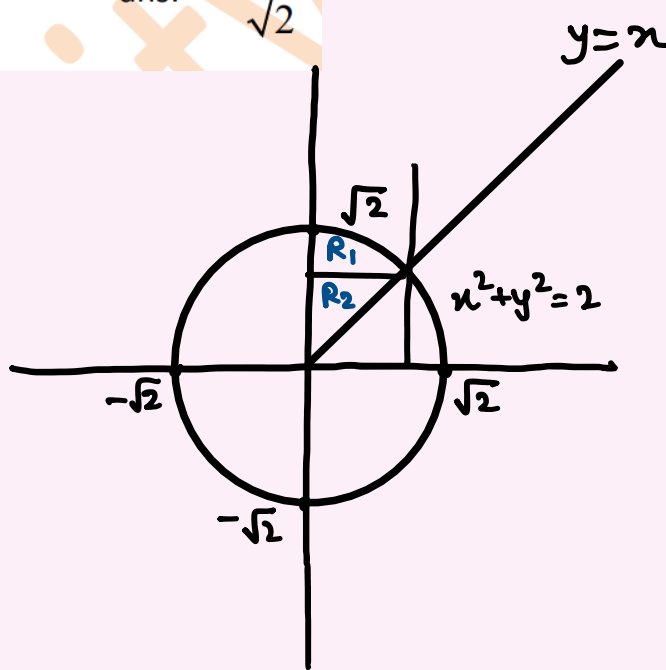
$$\int_0^1 \int_{y^2}^{2y^2} \frac{t^{-1/2}}{2} dt dy + \int_1^{\sqrt{2}} \int_{y^2}^2 \frac{t^{-1/2}}{2} dt dy$$

$$= \int_0^1 \left[\frac{t^{1/2}}{\frac{2}{2}} \right]_{y^2}^{2y^2} dy + \int_1^{\sqrt{2}} \left[\frac{t^{1/2}}{\frac{2}{2}} \right]_{y^2}^2 dy$$

$$= \int_0^1 (\sqrt{2y} - y) dy + \int_1^{\sqrt{2}} (\sqrt{2} - y) dy$$

$$= \left(\frac{\sqrt{2}}{2} y^2 - \frac{y^2}{2} \right)_0^1 + \left(\sqrt{2} y - \frac{y^2}{2} \right)_1^{\sqrt{2}}$$

$$= \frac{\sqrt{2}-1}{2} + 2 - 1 - \sqrt{2} + \frac{1}{2} = 1 - \frac{1}{\sqrt{2}}$$



Unit-1 class-6

$$1. \int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dx dy dz$$

$$\begin{aligned} A. & \int_{-c}^c \int_{-b}^b \left(\frac{x^3}{3} + y^2 x + z^2 x \right)_{-a}^a dy dz \\ &= \int_{-c}^c \int_{-b}^b \left(\frac{a^3}{3} + ay^2 + az^2 \right) - \left(-\frac{a^3}{3} - ay^2 - az^2 \right) dy dz \\ &= 2 \int_{-c}^c \int_{-b}^b \left(\frac{a^3}{3} + ay^2 + az^2 \right) dy dz \\ &= 2 \int_{-c}^c \left[\frac{a^3 y}{3} + \frac{ay^3}{3} + ayz^2 \right]_{-b}^b dz \\ &= 2 \int_{-c}^c \left(\frac{a^3(b+b)}{3} + \frac{a(b^3+b^3)}{3} + az^2(b+b) \right) dz \\ &= 4 \int_{-c}^c \left(\frac{a^3 b}{3} + \frac{ab^3}{3} + abz^2 \right) dz \\ &= 4 \left[\frac{a^3 bz}{3} + \frac{ab^3 z}{3} + \frac{abz^3}{3} \right]_{-c}^c \\ &= \frac{8abc}{3} (a^2 + b^2 + c^2) \end{aligned}$$

$$2. \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx \quad \text{ans: } \frac{1}{8}e^{4a} - \frac{3}{4}e^{2a} + e^a - \frac{3}{8}$$

$$\begin{aligned} A. & \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx \\ &= \int_0^a \int_0^x e^{x+y} (e^{x+y} - 1) dy dx \\ &= \int_0^a \int_0^x (e^{2x+2y} - e^{x+y}) dy dx \\ &= \int_0^a \left[\frac{e^{2x+2y}}{2} - (e^{x+y}) \right]_0^x dx \\ &= \int_0^a \left(\frac{e^{4x}}{2} - \frac{e^{2x}}{2} - e^{2x} + e^x \right) dx \\ &= \left[\frac{e^{4x}}{8} - \frac{e^{2x}}{4} - \frac{e^{2x}}{2} + e^x \right]_0^a \\ &= \left[\frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a \right]_0^a \\ &= \frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a - \left(\frac{1}{8} - \frac{3}{4} + 1 \right) \\ &= \frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a - \frac{3}{8} \end{aligned}$$

$$3. \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta$$

$$\text{ans: } \frac{5\pi a^3}{64}$$

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \left[rz \right]_0^{\frac{a^2-r^2}{a}} dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \left(\frac{ra^2 - r^3}{a} \right) dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{r^2 a}{2} - \frac{r^4}{4a} \right)_0^{a \sin \theta} d\theta \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{a^3 \sin^2 \theta}{2} - \frac{a^3 \sin^4 \theta}{4} \right) d\theta \\ &= \int_0^{\frac{\pi}{2}} \left[\frac{a^3}{2} \left(\frac{1 - \cos 2\theta}{2} \right) - \frac{a^3}{4} \left((\sin^2 \theta)^2 \right) \right] d\theta \\ &= \int_0^{\frac{\pi}{2}} \left[\frac{a^3}{2} \left(\frac{1 - \cos 2\theta}{2} \right) - \frac{a^3}{4} \left(\frac{1 - \cos 2\theta}{2} \right)^2 \right] d\theta \\ &= \int_0^{\frac{\pi}{2}} \left[\frac{a^3}{4} - \frac{a^3 \cos 2\theta}{4} - \frac{a^3}{16} \left(1 + \cos^2 2\theta - 2 \cos 2\theta \right) \right] d\theta \\ &= \int_0^{\frac{\pi}{2}} \left[\frac{a^3}{4} - \frac{a^3 \cos 2\theta}{4} - \frac{a^3}{16} - \frac{a^3}{16} \left(\frac{1 + \cos 4\theta}{2} \right) + \frac{a^3 \cos 2\theta}{8} \right] d\theta \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{a^3}{4} - \frac{a^3 \cos 2\theta}{4} - \frac{a^3}{16} - \frac{a^3}{32} - \frac{a^3 \cos 4\theta}{32} + \frac{a^3 \cos 2\theta}{8} \right) d\theta \\ &= \left[\frac{a^3 \theta}{4} - \frac{a^3 \sin 2\theta}{8} - \frac{a^3 \theta}{16} - \frac{a^3 \theta}{32} - \frac{a^3 \sin 4\theta}{128} + \frac{a^3 \sin 2\theta}{16} \right]_0^{\frac{\pi}{2}} \\ &= \frac{a^3 \pi}{8} - \frac{a^3}{8} (0) - \frac{a^3 \pi}{32} - \frac{a^3 \pi}{64} - \frac{a^3}{128} (0) + \frac{a^3}{16} (0) \\ &= \frac{8-2-1}{64} \times a^3 \pi = \frac{5\pi a^3}{64} \end{aligned}$$

Unit-1 class-7

1. Find the volume of the tetrahedron bounded by co-ordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
ans: $\frac{abc}{6}$

A. $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$\frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b} \Rightarrow z = \left(1 - \frac{x}{a} - \frac{y}{b}\right) \cdot c$

On x - y plane, $z=0$

$\frac{x}{a} + \frac{y}{b} = 1$

$y = \left(1 - \frac{x}{a}\right) \cdot b$

$z \Rightarrow 0$ to $\left(1 - \frac{x}{a} - \frac{y}{b}\right) \cdot c$

$y \Rightarrow 0$ to $\left(1 - \frac{x}{a}\right) \cdot b$

$x \Rightarrow 0$ to a
 $\Rightarrow \text{Volume} = \int_0^a \int_0^{\left(1 - \frac{x}{a}\right)b} \int_0^{\left(1 - \frac{x}{a} - \frac{y}{b}\right)c} dz dy dx$

$= \int_0^a \int_0^{\left(1 - \frac{x}{a}\right)b} \left[z \right]_0^{\left(1 - \frac{x}{a} - \frac{y}{b}\right)c} dy dx$

$= \int_0^a \int_0^{\left(1 - \frac{x}{a}\right)b} \left(1 - \frac{x}{a} - \frac{y}{b}\right)c dy dx$

$= \int_0^a \left[cy - \frac{cxy}{a} - \frac{cy^2}{2b} \right]_0^{\left(1 - \frac{x}{a}\right)b} dx$

$= \int_0^a \left[\left(c - \frac{cx}{a}\right)\left(1 - \frac{x}{a}\right)b - \frac{c}{2b}\left(1 - \frac{x}{a}\right)^2 b^2 \right] dx$

$= \int_0^a \left[\left(1 - \frac{x}{a}\right)^2 bc - \frac{bc}{2}\left(1 - \frac{x}{a}\right)^2 \right] dx$

$= \int_0^a \frac{bc}{2} \left(1 - \frac{x}{a}\right)^2 dx$

$= \frac{bc}{2} \int_0^a \left(1 + \frac{x^2}{a^2} - \frac{2x}{a}\right) dx = \frac{bc}{2} \left(x + \frac{x^3}{3a^2} - \frac{x^2}{a} \right)_0^a$

$= \frac{bc}{2} \left(a - a + \frac{a^3}{3a^2} \right) = \frac{abc}{6}$

2. Find the volume of the region bounded by $z = x^2 + y^2, z=0, x=-a, x=a, y=-a, y=a$
ans: $\frac{8a^4}{3}$

$z \rightarrow 0$ to $x^2 + y^2$

$y \rightarrow -a$ to a

$x \rightarrow -a$ to a

$\int_{-a}^a \int_{-a}^a \int_0^{x^2+y^2} dz dy dx = \int_{-a}^a \int_{-a}^a (x^2 + y^2) dy dx$

$= \int_{-a}^a \left(x^2 y + \frac{y^3}{3} \right)_{-a}^a dx = \int_{-a}^a \left(ax^2 + \frac{a^3}{3} - \left(-ax^2 - \frac{a^3}{3} \right) \right) dx$

$= 2 \int_{-a}^a \left(ax^2 + \frac{a^3}{3} \right) dx = 2 \left(\frac{ax^3}{3} + \frac{a^3 x}{3} \right)_{-a}^a$

$= 2 \left(\frac{a^4}{3} + \frac{a^4}{3} - \left(-\frac{a^4}{3} - \frac{a^4}{3} \right) \right)$

$= 2 \left(\frac{4a^4}{3} \right) = \frac{8a^4}{3}$

3. Find the volume cut off from the cylinder $x^2 + y^2 = ax$ by the planes $z=0$ and $z=x$ ans: $\frac{\pi a^3}{8}$

$x^2 + y^2 = ax \Rightarrow y = \sqrt{ax - x^2}$

$z \Rightarrow 0$ to x

$y \Rightarrow -\sqrt{ax - x^2}$ to $\sqrt{ax - x^2}$

$x \Rightarrow 0$ to a

$\int_0^a \int_{-\sqrt{ax-x^2}}^{\sqrt{ax-x^2}} \int_0^x dz dy dx$

$\Rightarrow \int_0^a \int_{-\sqrt{ax-x^2}}^{\sqrt{ax-x^2}} x dy dx = \int_0^a 2x \sqrt{ax - x^2} dx = 2 \int_0^a \sqrt{ax^3 - x^4} dx$

$= 2 \int_0^a x^{3/2} \sqrt{a - x} dx$

$u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}} dx \Rightarrow 2 du \cdot u = dx$

$\Rightarrow 2 \int_0^{\sqrt{a}} u^3 \sqrt{a - u^2} \cdot 2u du = 4 \int_0^{\sqrt{a}} u^4 \sqrt{a - u^2} du$

$u = \sqrt{a} \sin t \quad du = \sqrt{a} \cos t dt$

$u = \sqrt{a} \Rightarrow t = \pi/2$

$u = 0 \Rightarrow t = 0$

$\Rightarrow 4 \int_0^{\pi/2} a^2 \sin^4 t \sqrt{a(1 - \sin^2 t)} \cdot \sqrt{a} \cos t dt$

$= 4 \int_0^{\pi/2} a^3 \sin^4 t \cos^3 t dt$

$= 4 \int_0^{\pi/2} a^3 \sin^4 t (1 - \sin^2 t) dt$

$= 4a^3 \int_0^{\pi/2} (\sin^4 t - \sin^6 t) dt$

$= 4a^3 \int_0^{\pi/2} \left(\left(\frac{1 + \cos 2t}{2} \right)^2 - \left(\frac{1 + \cos 2t}{2} \right)^3 \right) dt$

$= 4a^3 \int_0^{\pi/2} \left(\frac{1 + \cos^2 2t + 2\cos 2t}{4} - \frac{(1 + 3\cos^2 2t + 3\cos 2t + \cos^3 2t)}{8} \right) dt$

$= \frac{4a^3}{8} \int_0^{\pi/2} \left[2 + 2\left(1 + \frac{\cos 4t}{2}\right) + 4\cos 2t - 1 - 3\left(1 + \frac{\cos 4t}{2}\right) - 3\cos 2t - \cos^3 2t \right] dt$

$= \frac{a^3}{2} \int_0^{\pi/2} \left[\frac{1}{2} - \frac{\cos 4t}{2} + \cos 2t - \cos 2t - \cos^3 2t \right] dt$

$= \frac{a^3}{2} \left[\frac{t}{2} - \frac{\sin 4t}{8} + \frac{\sin 2t}{2} - \frac{\sin 2t}{2} - \frac{\sin^3 2t}{6} \right]_0^{\pi/2}$

$= \frac{a^3}{2} \times \left[\frac{\pi}{2} - 0 \right] = \frac{\pi a^3}{8}$

Unit-1 class-8

1. Find the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$ ans: $\frac{16a^3}{3}$

y varies from 0 to $\sqrt{a^2 - x^2}$
 z varies from 0 to $\sqrt{a^2 - x^2}$
 x varies from 0 to a

$$8 \int_0^a \int_0^{\sqrt{a^2 - x^2}} \int_0^{\sqrt{a^2 - x^2}} dz dy dx$$

$$= 8 \int_0^a \int_0^{\sqrt{a^2 - x^2}} [z]_0^{\sqrt{a^2 - x^2}} dy dx$$

$$= 8 \int_0^a (a^2 - x^2) [y]_0^{\sqrt{a^2 - x^2}} dx$$

$$= 8 \int_0^a (a^2 - x^2) dx = 8 \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \left(a^3 - \frac{a^3}{3} \right) \times 8 = \frac{16a^3}{3}$$

2. Find the volume cut from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$ above XY plane.

ans: $\frac{\pi a^3}{3} (2 - \sqrt{2})$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r \rightarrow 0 \text{ to } a$$

$$\phi \rightarrow 0 \text{ to } 2\pi$$

$$\theta \rightarrow 0 \text{ to } \pi/4$$

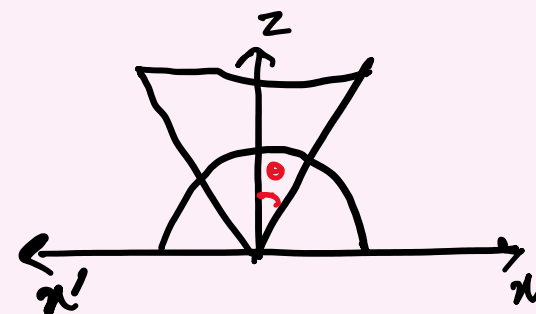
$$\int_0^{2\pi} \int_0^{\pi/4} \int_0^a r^2 \sin \theta dr d\theta d\phi$$

$$= \int_0^{2\pi} \int_0^{\pi/4} \left[\frac{r^3}{3} \right]_0^a \sin \theta d\theta d\phi$$

$$= \frac{a^3}{3} (-\cos \theta)_0^{\pi/4} \times 2\pi$$

$$= \frac{\pi a^3}{3} \left(-\frac{1}{\sqrt{2}} + 1 \right) \times 2$$

$$= \frac{\pi a^3}{3} (2 - \sqrt{2})$$



$$x^2 + y^2 = z^2$$

$$x^2 + y^2 + z^2 = 2z^2$$

$$r^2 = 2z^2 \cos^2 \theta$$

$$\cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$$

3. Find the average value of $f(x, y, z) = x + y + z$, using triple integrals over the region $D = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq 3, 0 \leq z \leq 5\}$. ans: $\frac{9}{2}$

$$\text{Avg} = \frac{\int_0^1 \int_0^3 \int_0^5 (x + y + z) dz dy dx}{\int_0^1 \int_0^3 \int_0^5 dz dy dx}$$

$$\text{Num} \Rightarrow \int_0^1 \int_0^3 \left(xz + yz + \frac{z^2}{2} \right)_0^5 dy dx$$

$$= \int_0^1 \int_0^3 \left(5x + 5y + \frac{25}{2} \right) dy dx$$

$$= \int_0^1 \left(5xy + \frac{5y^2}{2} + \frac{25y}{2} \right)_0^3 dx$$

$$= \int_0^1 \left(15x + \frac{45}{2} + \frac{75}{2} \right) dx$$

$$= \left[\frac{15x^2}{2} + 60x \right]_0^1 = \frac{135}{2}$$

$$\text{Avg} = \frac{135}{2 \times 15} = \frac{9}{2}$$

$$\text{Den: } \int_0^1 \int_0^3 \int_0^5 dz dy dx$$

$$= 5 \times 3 \times 1$$

$$= 15$$

Unit-1 class-9

1. evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$ by changing to spherical coordinates. Ans: $\frac{\pi^2}{8}$

$$\left. \begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned} \right\} J = r^2 \sin \theta$$

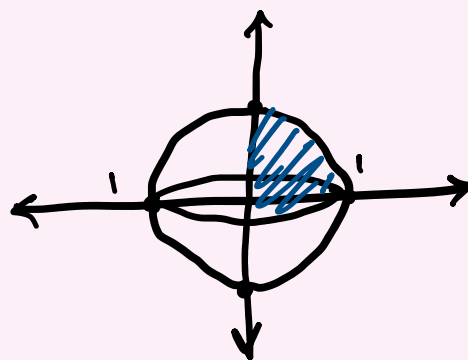
$$x^2 + y^2 + z^2 = r^2$$

$$\int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \frac{r^2 \sin \theta dr d\theta d\phi}{\sqrt{1-r^2}}$$

$$\int_0^{\pi/2} \int_0^{\pi/2} \int_0^{\pi/2} \frac{\sin^2 t \sin \theta \cos t dt d\theta d\phi}{\cos t}$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^{\pi/2} \left(\frac{1 - \cos 2t}{2} \right) \sin \theta dt d\theta d\phi$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \left(\frac{t}{2} - \frac{\sin 2t}{4} \right)_0^{\pi/2} \sin \theta d\theta d\phi = \frac{\pi}{4} \times (-\cos \theta)_0^{\pi/2} \times \frac{\pi}{2} = \frac{\pi^2}{8}$$



$$r = \sin t \\ dr = \cos t dt$$

2. Evaluate $\iiint_V \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$ where V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$ ans: $2\pi a^2$

$$r \Rightarrow 0 \text{ to } a$$

$$\theta \Rightarrow 0 \text{ to } 2\pi$$

$$\phi \Rightarrow 0 \text{ to } 2\pi$$

$$\int_0^{2\pi} \int_0^{2\pi} \int_0^a \frac{r^2 \sin \theta dr d\theta d\phi}{\sqrt{r^2}} = \int_0^{2\pi} \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^a \sin \theta d\theta d\phi$$

$$= \frac{a^2}{2} \int_0^{2\pi} [-\cos \theta]_0^{2\pi} d\phi = \frac{a^2}{2} [2] \times 2\pi = 2\pi a^2$$

Unit-1 class-10

1. If R is the region bounded by $x = 0, y = 0, z = 0, z = 1$ and the cylinder $x^2 + y^2 = 1$ evaluate $\iiint xyz dx dy dz$ by changing to cylindrical coordinates. Ans: $1/16$

$$x = r \cos \theta ; y = r \sin \theta ; z = z$$

$$z \rightarrow 0 \text{ to } 1$$

$$r \rightarrow 0 \text{ to } 1$$

$$\theta \rightarrow 0 \text{ to } \pi/2$$

$$\int_0^{\pi/2} \int_0^1 \int_0^1 r^2 \cos \theta \sin \theta x z dz dr d\theta$$

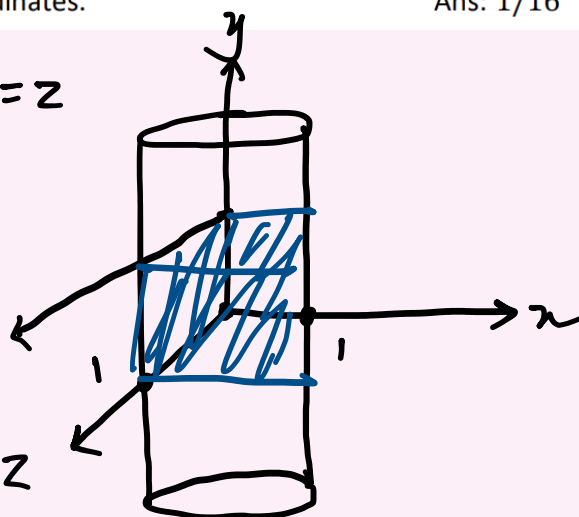
$$= \int_0^{\pi/2} \int_0^1 r^3 \left[\frac{z^2}{2} \right]_0^1 \cos \theta \sin \theta dr d\theta = \frac{1}{2} \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_0^1 \cos \theta \sin \theta d\theta$$

$$t = \sin \theta \Rightarrow dt = \cos \theta d\theta$$

$$\theta = 0 \Rightarrow t = 0$$

$$\theta = \pi/2 \Rightarrow t = 1$$

$$= \int_0^1 \frac{1}{8} x t dt = \left[\frac{t^2}{2} \right]_0^1 \times \frac{1}{8} = \frac{1}{16}$$



2. Evaluate $\iiint (x^2 + y^2) dx dy dz$ over the region bounded by the paraboloid $x^2 + y^2 = 3z$ and the plane $z = 3$ ans: $\frac{81\pi}{2}$

$$x = r \cos \theta ; y = r \sin \theta$$

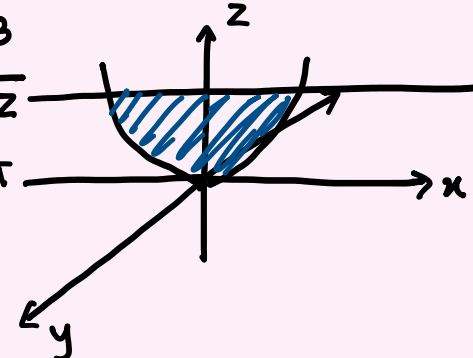
$$x^2 + y^2 = r^2$$

$$dx dy dz = r dz dr d\theta$$

$$z \Rightarrow 0 \text{ to } 3$$

$$r \Rightarrow 0 \text{ to } \sqrt{3z}$$

$$\theta \Rightarrow 0 \text{ to } 2\pi$$



$$\int_0^{2\pi} \int_0^3 \int_0^{\sqrt{3z}} r^2 \cdot r dr dz d\theta$$

$$= \int_0^{2\pi} \int_0^3 \left[\frac{r^4}{4} \right]_0^{\sqrt{3z}} dz d\theta = \int_0^{2\pi} \int_0^3 \frac{9z^2}{4} dz d\theta$$

$$= \int_0^{2\pi} \left[\frac{9z^3}{12} \right]_0^3 d\theta = \int_0^{2\pi} \frac{9 \times 27}{4} d\theta = \frac{81}{4} \times 2\pi = \frac{81\pi}{2}$$

3. Evaluate $\iiint \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$ over the region bounded by the sphere $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$ $a > b > 0$ ans: $2\pi(a^2 - b^2)$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$x^2 + y^2 + z^2 = r^2$$

$$r \rightarrow b \text{ to } a$$

$$\theta \rightarrow 0 \text{ to } \pi$$

$$\phi \rightarrow 0 \text{ to } 2\pi$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

$$\int_0^{2\pi} \int_0^\pi \int_b^a \frac{r^2 \sin \theta dr d\theta d\phi}{r} = \int_0^{2\pi} \int_0^\pi \left[\frac{r^2}{2} \right]_b^a \sin \theta d\theta d\phi$$

$$= \frac{a^2 - b^2}{2} \int_0^{2\pi} (-\cos \theta)_0^\pi d\phi = (a^2 - b^2) \pi (1 + 1) = 2\pi(a^2 - b^2)$$

1. A lamina is bounded by the curves $y = x^2 - 3x$ and $y = 2x$. If the density at any point is given by λxy , find by double integration, mass of the lamina. Ans: $182\frac{7}{24}\lambda$

$y = x^2 - 3x$
 $y = 2x$
common point $\Rightarrow x = 0, 5$

$y = x^2 - 3x$
 $x = 0 \Rightarrow y = 0$
 $x = 1 \Rightarrow y = -2$
 $x = 2 \Rightarrow y = -2$
 $x = 3 \Rightarrow y = 0$
 $x = -1 \Rightarrow y = 4$
 $x = 4 \Rightarrow y = 4$

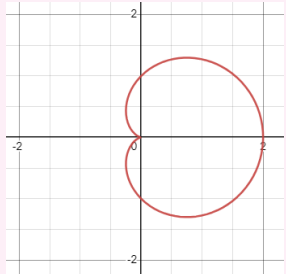
$y = 2x$
 $x = 0 \Rightarrow y = 0$
 $x = 1 \Rightarrow y = 2$
 $x = 2 \Rightarrow y = 4$

Line $\parallel$ to y-axis,
 $y \Rightarrow x^2 - 3x$ to $2x$
 $x \Rightarrow 0$ to 5

$$\int_0^5 \int_{x^2-3x}^{2x} \lambda xy \, dy \, dx$$
$$\int_0^5 \left[\frac{\lambda xy^2}{2} \right]_{x^2-3x}^{2x} dx$$
$$= \int_0^5 \frac{\lambda x}{2} (2x)^2 - (x^2 - 3x)^2 dx = \int_0^5 \frac{\lambda x}{2} (4x^2 - (x^4 - 6x^3 + 9x^2)) dx$$
$$= \frac{\lambda}{2} \int_0^5 (4x^3 - x^4 + 6x^3 - 9x^2) dx = \frac{\lambda}{2} \left[\frac{4x^4}{4} - \frac{x^5}{5} + \frac{6x^4}{4} - \frac{9x^3}{3} \right]_0^5$$
$$= \frac{4375\lambda}{24} = 182\frac{7}{24}\lambda$$

2. Find the mass of the lamina in the form of the cardioid $r = a(1 + \cos \theta)$ whose density at any point varies as the square of its distance from the initial line. ans: $\frac{21\pi a^4 \mu^4}{32}$

$r = a(1 + \cos \theta)$
 $\rho \propto r^2 \sin^2 \theta \Rightarrow \rho = \mu r^2 \sin^2 \theta$
 $r^2 = x^2 + y^2$ where $x = r \cos \theta$
 $y = r \sin \theta$



$dm = \rho \, r \, dr \, d\theta$
 $r \Rightarrow 0$ to $a(1 + \cos \theta)$
 $\theta \Rightarrow 0$ to 2π

$$\iint \rho \cdot r \, dr \, d\theta = \int_0^{2\pi} \int_0^{a(1+\cos \theta)} \mu r^3 \sin^2 \theta \, dr \, d\theta = \int_0^{2\pi} \left[\frac{\mu r^4}{4} \right]_0^{a(1+\cos \theta)} \sin^2 \theta \, d\theta$$
$$= \int_0^{2\pi} \frac{\mu a^4}{4} (1 + \cos \theta)^4 \sin^2 \theta \, d\theta$$
$$= \frac{2 \times \mu a^4}{4} \int_0^\pi (1 + \cos \theta)^4 \sin^2 \theta \, d\theta$$
$$= \frac{\mu a^4}{2} \int_0^\pi \left[2 \cos^2 \frac{\theta}{2} \right]^4 \times 4 \sin^2 \frac{\theta}{2} \cdot \cos^2 \frac{\theta}{2} \, d\theta$$
$$= \frac{\mu a^4}{2} \times 16 \times \frac{\pi}{4} \int_0^\pi \sin^2 \frac{\theta}{2} \cdot \cos^{10} \frac{\theta}{2} \, d\theta$$
$$= 32 \mu a^4 \int_0^\pi \sin^2 \frac{\theta}{2} \cdot \cos^{10} \frac{\theta}{2} \, d\theta$$

$t = \frac{\theta}{2} \Rightarrow 2dt = d\theta$
 $= 32 \mu a^4 \int_0^{\pi/2} \sin^2 t \cdot \cos^{10} t \cdot 2dt$
 $= 64 \mu a^4 \times \frac{1}{2} B\left(\frac{3}{2}, \frac{11}{2}\right)$
 $= 32 \mu a^4 \times \frac{\left[\frac{3}{2}\right] \left[\frac{11}{2}\right]}{\sqrt{\pi}} = 32 \mu a^4 \times \frac{\frac{\sqrt{\pi}}{2} \times \left(\frac{7 \times 5 \times 3 \times 1 \times \sqrt{\pi}}{2 \times 2 \times 2 \times 2 \times 2}\right)}{\frac{6 \times 5 \times 4 \times 3 \times 2}{2}}$
 $= \frac{21\pi \mu a^4}{32}$

3. Find the mass of a solid in the form of the positive octant of the sphere $x^2 + y^2 + z^2 = 9$ if the density at any point is $2xyz$ ans: 30.375

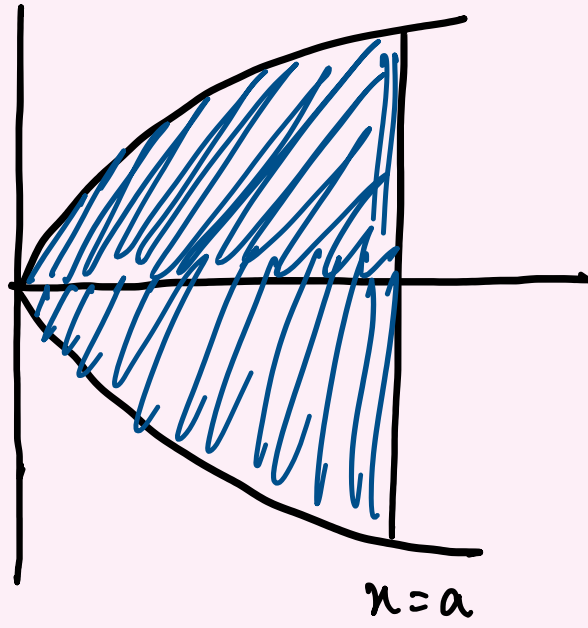
$\rho = 2xyz$
 $z \Rightarrow 0$ to $\sqrt{9 - y^2 - x^2}$
 $y \Rightarrow 0$ to $\sqrt{9 - x^2}$
 $x \Rightarrow 0$ to 3

$$m = \int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-y^2-x^2}} 2xyz \cdot dz \, dy \, dx$$
$$= \int_0^3 \int_0^{\sqrt{9-x^2}} \left[xyz^2 \right]_0^{\sqrt{9-y^2-x^2}} dy \, dx = \int_0^3 \int_0^{\sqrt{9-x^2}} (xy(9 - x^2 - y^2)) dy \, dx$$
$$= \int_0^3 \int_0^{\sqrt{9-x^2}} (9xy - x^3y - xy^3) dy = \int_0^3 \left[\frac{9xy^2}{2} - \frac{x^3y^2}{2} - \frac{xy^4}{4} \right]_0^{\sqrt{9-x^2}} dx$$
$$= \int_0^3 \left[\frac{9x(9-x^2)}{2} - \frac{x^3(9-x^2)}{2} - \frac{x(9-x^2)^2}{4} \right] dx$$
$$= \int_0^3 \frac{162x - 18x^3 - 18x^3 + 2x^5 - 81x - x^5 + 18x^3}{4} dx$$
$$= \int_0^3 \left(\frac{x^5 - 18x^3 + 81x}{4} \right) dx = \left[\frac{x^6}{24} - \frac{9x^4}{8} + \frac{81x^2}{8} \right]_0^3 = \frac{243}{8}$$

4. Find the centroid of the area enclosed by the parabola $y^2 = 4ax$, the axis of x and its latus rectum. Ans: $\left(\frac{3a}{20}, \frac{3a}{16}\right)$

No answer found, please contact me if you do find one

(8296465473)



Unit-1 class-12

1. Using double integrals find moment of inertia about the x-axis of the area enclosed by the lines

$$x=0, y=0, \frac{x}{a} + \frac{y}{b} = 1$$

$$\text{Ans: } \frac{ab^3}{12}$$

2. Find the moment of inertia of an octant of ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ about x-axis.

$$\text{Ans: } \frac{abc(b^2 + c^2)\pi}{30}$$

$$1. \Rightarrow \frac{x}{a} + \frac{y}{b} = 1$$

$$\int_0^a \int_0^{b(1-\frac{x}{a})} y^2 dy dx = \int_0^a \left[\frac{y^3}{3} \right]_0^{b(1-\frac{x}{a})} dx$$

$$\frac{1}{3} \int_0^a b^3 \left(1 - \frac{x}{a}\right)^3 dx = \frac{1}{3} \int_0^a \frac{b^3 (a-x)^3}{a^3} dx = \frac{b^3}{3a^3} \int_0^a (a-x)^3 dx$$

$$= \frac{b^3}{3a^3} \int_0^a (a^3 - x^3 - 3a^2x + 3ax^2) dx = \frac{b^3}{3a^3} \left[a^3x - \frac{x^4}{4} - \frac{3a^2x^2}{2} + \frac{3ax^3}{3} \right]$$

$$= \frac{b^3}{3a^3} \left[a^4 - \frac{a^4}{4} - \frac{3a^4}{2} + \frac{3a^4}{3} \right] = \frac{a^4 b^3}{12a^3} = \frac{ab^3}{12}$$

$$2. \text{ MOI} = \iiint \rho(x, y, z) (z^2 + y^2) dx dy dz$$

$$x = ar \sin \theta \cos \phi$$

$$y = br \sin \theta \sin \phi$$

$$z = cr \cos \theta$$

$$dx dy dz = abc r^2 \sin \theta dr d\theta d\phi$$

$$r = 0 \text{ to } 1$$

$$\theta = 0 \text{ to } \pi/2$$

$$\phi = 0 \text{ to } \pi/2$$

$$\text{Now, MOI} = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 (b^2 r^2 \sin^2 \theta \sin^2 \phi + c^2 r^2 \cos^2 \theta) abc r^2 \sin \theta dr d\theta d\phi$$

$$= abc \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 (b^2 r^4 \sin^3 \theta \sin^2 \phi + c^2 r^4 \sin \theta \cos^2 \theta) dr d\theta d\phi$$

$$= \frac{abc}{5} \int_0^{\pi/2} \int_0^{\pi/2} (b^2 \sin^3 \theta \sin^2 \phi + c^2 \sin \theta \cos^2 \theta) d\theta d\phi$$

$$= \frac{abc}{5} \left[\frac{b^2 \pi}{6} + \frac{c^2 \pi}{6} \right] = \frac{abc \pi (b^2 + c^2)}{30}$$